

Projet : Bioinformatique des séquences

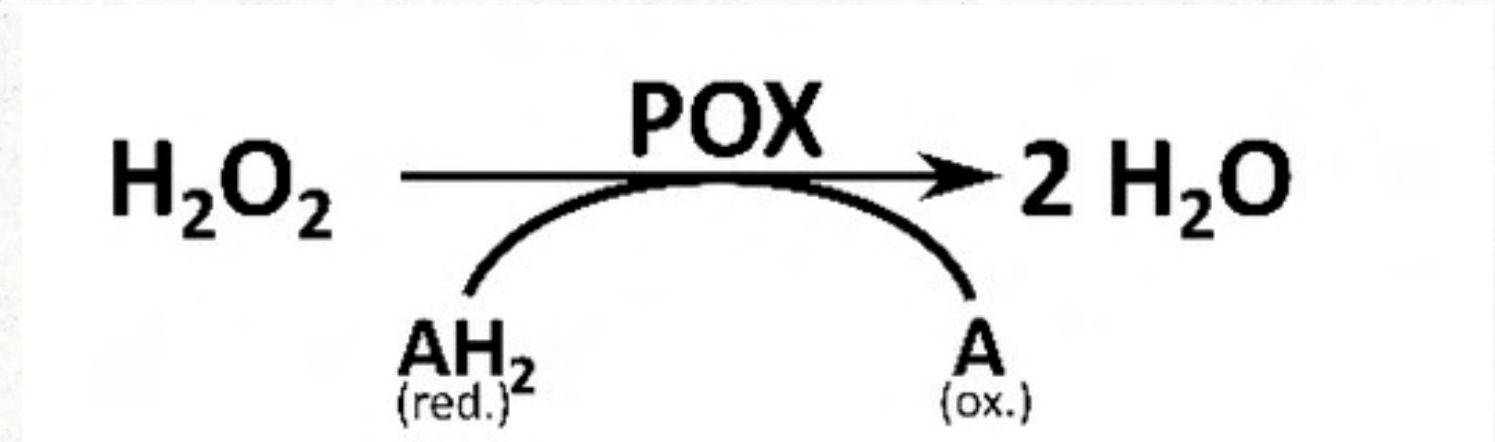
Caractérisation de la famille des peroxydases de classe III

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MASTER BIOINFORMATIQUE ET BIOLOGIE DES SYSTÈMES



Que sont les peroxydases ?

- Enzymes
- Universelles du monde vivant
- Utilisent le peroxyde d'hydrogène (H_2O_2) → oxyder leur substrat (réduction H_2O_2 en H_2O)





DESCRIPTION et CARACTERISATION

Les peroxydases à Hème (Superfamille)



CLASSE I

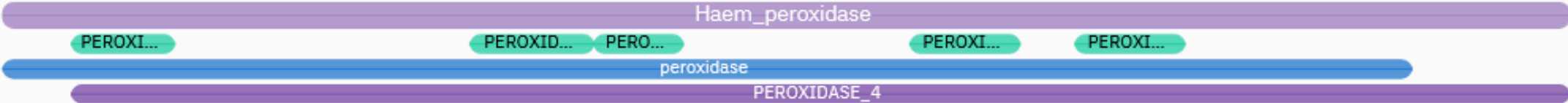
▼ Representative Domains



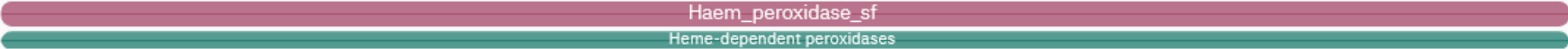
▼ Family



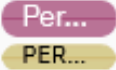
▼ Domain



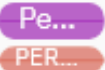
▼ Homologous Superfamily



▼ Active Site



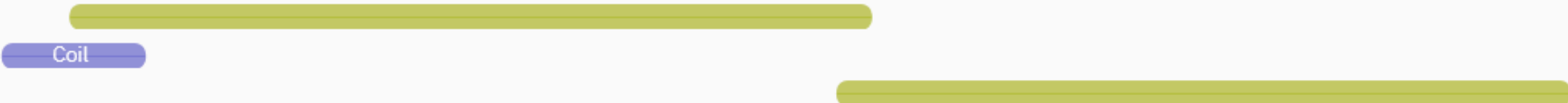
▼ Binding Site



▼ Unintegrated



▼ Other Features



This result was part of a multiple (99) sequence job. You can choose another sequence below:

◀ 2468[SoAPx03]

2491[McAPx01] ▶

Title	2468[SoAPx03]	
Job ID	iprscan5-R20241003-230004-0298-80054867-p1m-1	
Length	250 amino acids	
Actions		
Status	finished	
Expires	Fri Oct 11 2024	

Protein family membership

- ▼ [Heme-binding peroxidase Ccp1-like](#) (IPR044831)
- [Class I peroxidase](#) (IPR002207)

[Peroxidase_I - IPR002207](#)
ASPEROXIDASE - PR00459

[Ccp1-like - IPR044831](#)
THYLAKOID LUMENAL 29 KDA PROTEIN, CHLOROP

[Haem_peroxidase - IPR002016](#)
PEROXIDASE - PR00458
peroxidase - PF00141
PEROXIDASE_4 - PS50873

[Haem_peroxidase_sf - IPR010255](#)
Heme-dependent peroxidases - SSF48113

[Peroxidases_AS - IPR019794](#)
PEROXIDASE_2 - PS00436

[Peroxidases_heam-ligand_BS - IPR01979](#)
PEROXIDASE_1 - PS00435

G3DSA:1.10.520.10
Peroxidase, domain 2 - G3DSA:1.10.420.10
ascorbate_peroxidase - cd00691

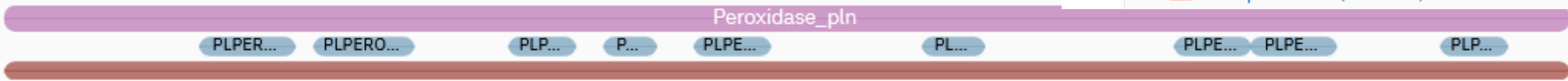
FUNFAM: G3DSA:1.10.520.10:FF:000003
COILS: Coil
FUNFAM: G3DSA:1.10.420.10:FF:000003

CLASSE III

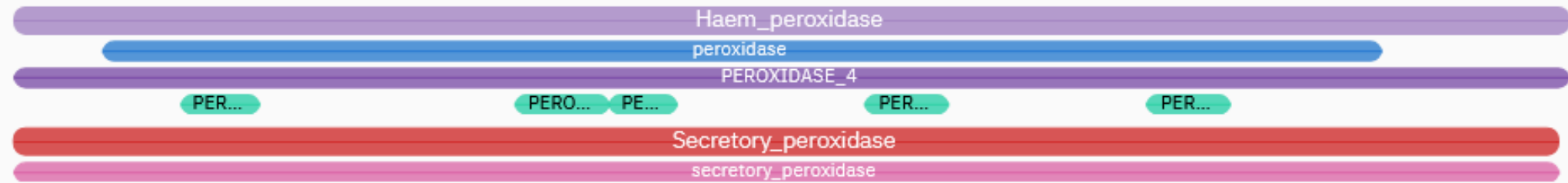
Representative Domains



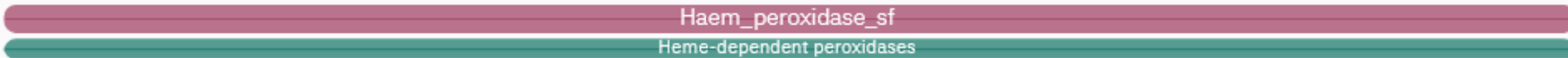
Family



Domain



Homologous Superfamily



Active Site



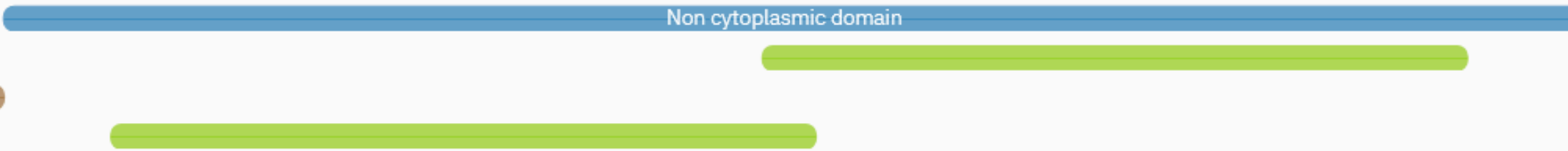
Binding Site



Unintegrated



Other Features



This result was part of a multiple (35) sequence job. You can choose another sequence below:

7808[BdiPrx11] [Brachypodium_distachyon]]Monocotyledons[Class_III_peroxidase]complete|

86[AtPrx10] [Arabidopsis_thaliana]]Eudicotyledons[Class_III]▶

Title	7808[BdiPrx11] [Brachypodium_distachyon]]Monocotyledons[Class_III_peroxidase]complete
Job ID ⓘ	iprsan5-R20241003-222305-0374-38213845-p1m-1 📋
Length	367 amino acids
Actions	🗑️
Status	✅ saved in browser

Protein family membership

F Plant peroxidase (IPR000823)

F Peroxi
PEROXIDA
PLPEROXID

D Haem_
PEROXIDA
peroxidase
PEROXIDA

D Secret
secretory_p

H Haem_
Heme-dep

S Peroxi
PEROXIDA

S Peroxi
PEROXIDA

G3DSA:1.
Peroxidas

PHOBIUS
FUNFAM:
PHOBIUS
FUNFAM:

Quels peuvent être les **outils** efficaces nous permettant de **distinguer** efficacement les peroxydases de **classe I** et de **classe III** ?

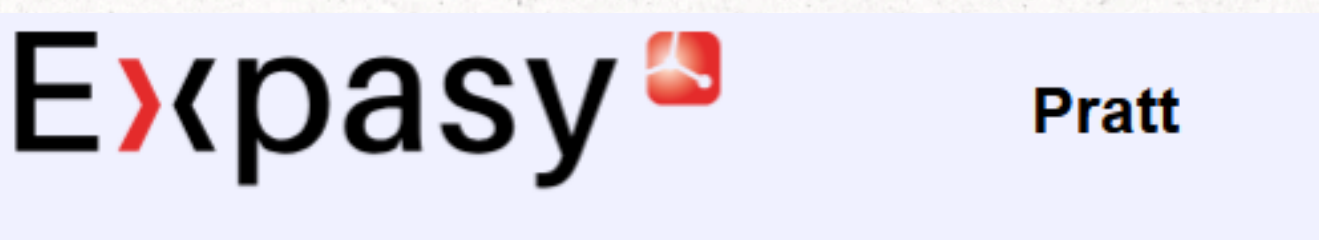
Multiple Sequence Alignment (MSA)



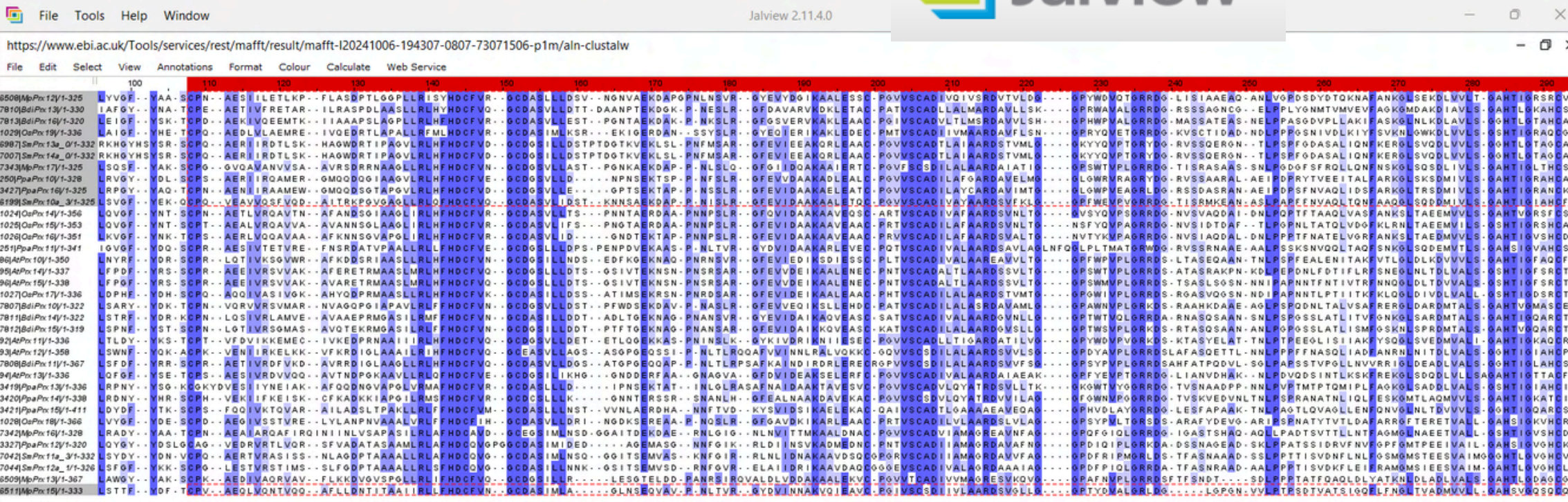
↪ **Cystéines conservées uniquement dans les séquences de la classe III**

Démarche : Signatures des domaines conservés

02



- Récupération des zones de similarité sur l'alignement multiple



Signatures : outil non suffisamment précis

→ pas de discrimination efficace des classes I et III

	Classe I	Classe III
Région issue de l'alignement multiple	225-328	110-290
Résultat signature PRATT	Pas de signature	R-[FILM]-x(2)-H-D-C-x(3,5)-G-C-[DE]-[ACG]-S-[ILV]-[LM]-[IL]
Comparaison signature VS SwissProt		96 séquences classe III

Démarche : Construction de profils HMM

- Extraction des zones de similarité sur l'alignement multiple : Script Bash

Classe I : 225-328

Classe III : 110-290

03



```
#!/bin/bash
awk '{
    if (NR == 1) {
        print;
        next;
    }
    seq = "";
    for (i = 1; i <= NF; i++) {
        if (i >= 110 && i <= 290) {
            seq = seq $i;
        } else if (i == 1) {
            header = $i;
        }
    }
    print header;
    print seq;
}' PB_classIII.mafft > PB_classIII_conserved_region_110_290.mafft
```

Classe III

```
#!/bin/bash
awk '{
    if (NR == 1) {
        print;
        next;
    }
    seq = "";
    for (i = 1; i <= NF; i++) {
        if (i >= 225 && i <= 328) {
            seq = seq $i;
        } else if (i == 1) {
            header = $i;
        }
    }
    print header;
    print seq;
}' /home/fmt017/work/projet_bioseq/alignement/mafft/PB_APx.mafft > PB_APx_region_225_328.mafft
```

Classe I

Démarche : Construction de profils HMM

Génération de HMM



```
Q=APX region 225_328
R=
S=
T=
U=
V=
W=
X=
Y=
Z=
[...]
```

HMM contre UniProt

E-value = probabilité que le score de correspondance entre le domaine/séquence et la HMM soit observée au hasard

Le score = donne la qualité de l'alignement entre la séquence cible et le modèle HMM.

Classe I 225-328

```
# Copyright (C) 2018 Howard Hughes Medical Institute.
# Freely distributed under the BSD open source license.
#
# query HMM file: PB_APx_region_225_328.hmm
# target sequence database: /bank/ebi/uniprot/current/fasta/uniprot_sprot.fasta
# output directed to file: PB_APx_region_225_328.search
#
```

Query: PB_APx_region_225_328 [M=526]									
Scores for complete sequences (score includes all domains):									
--- full sequence ---									
E-value	score	bias	E-value	score	bias	exp	N	Sequence	Description
3.4e-138	466.4	0.0	4.6e-138	466.0	0.0	1.2	1	sp Q69SV0 APX8_ORYSJ	Probable L-ascorbate peroxidase 8, chl
3.5e-136	459.8	0.0	3.8e-136	459.7	0.0	1.0	1	sp Q42564 APX3_ARATH	L-ascorbate peroxidase 3 OS=Arabidopsi
1.1e-135	458.1	0.3	1.4e-135	457.8	0.3	1.1	1	sp Q7XJ02 APX7_ORYSJ	Probable L-ascorbate peroxidase 7, chl
1.3e-135	457.9	0.0	1.5e-135	457.7	0.0	1.0	1	sp Q6ZJJ1 APX4_ORYSJ	Probable L-ascorbate peroxidase 4, per
3e-135	456.7	0.0	3.7e-135	456.4	0.0	1.0	1	sp Q42593 APXT_ARATH	L-ascorbate peroxidase T, chloroplasti
1.7e-134	454.2	0.0	2e-134	454.0	0.0	1.0	1	sp Q0JEQ2 APX3_ORYSJ	Probable L-ascorbate peroxidase 3, per
3.9e-134	453.0	0.0	4.7e-134	452.8	0.0	1.0	1	sp Q01MI9 APX3_ORYSI	Probable L-ascorbate peroxidase 3 OS=0
2.4e-133	450.4	0.0	2.6e-133	450.3	0.0	1.0	1	sp Q42592 APXS_ARATH	L-ascorbate peroxidase S, chloroplasti
2.6e-131	443.7	0.0	2.7e-131	443.6	0.0	1.0	1	sp P0C0L1 APX6_ORYSJ	Probable L-ascorbate peroxidase 6, chl
1.3e-129	438.1	0.0	1.5e-129	437.9	0.0	1.0	1	sp Q1PER6 APX2_ARATH	L-ascorbate peroxidase 2, cytosolic OS
1.1e-128	435.0	0.0	1.3e-128	434.8	0.0	1.0	1	sp Q7XZP5 APX5_ARATH	L-ascorbate peroxidase 5, peroxisomal

Classe III 110-290

```
# HMMSEARCH v1. search profile(s) against a sequence database
# HMMER 3.2.1 (June 2018); http://hmmer.org/
# Copyright (C) 2018 Howard Hughes Medical Institute.
# Freely distributed under the BSD open source license.
#
# query HMM file: PB_classIII_conserved_region_110_290.hmm
# target sequence database: /bank/ebi/uniprot/current/fasta/uniprot_sprot.fasta
# output directed to file: PB_classIII_conserved_region_110_290.search
#
```

Query: PB_classIII_conserved_region_110_290 [M=436]									
Scores for complete sequences (score includes all domains):									
--- full sequence ---									
E-value	score	bias	E-value	score	bias	exp	N	Sequence	Description
4.6e-147	495.3	1.1	5.1e-147	495.1	1.1	1.0	1	sp Q23237 PER49_ARATH	Peroxidase 49 OS=Arabidopsis thaliana
4.9e-146	491.9	0.6	5.4e-146	491.7	0.6	1.0	1	sp Q9SI16 PER15_ARATH	Peroxidase 15 OS=Arabidopsis thaliana
1.1e-145	490.7	1.4	1.2e-145	490.6	1.4	1.0	1	sp A7NY33 PER4_VITVI	Peroxidase 4 OS=Vitis vinifera OX=2976
3e-144	486.0	2.7	3.3e-144	485.8	2.7	1.0	1	sp P22195 PER1_ARAHY	Cationic peroxidase 1 OS=Arachis hypog
1.4e-143	483.8	0.0	1.5e-143	483.6	0.0	1.0	1	sp Q96512 PER9_ARATH	Peroxidase 9 OS=Arabidopsis thaliana
3e-143	482.7	0.1	3.3e-143	482.6	0.1	1.0	1	sp A7QEU4 PER5_VITVI	Peroxidase 5 OS=Vitis vinifera OX=2976
1.4e-142	480.5	0.6	1.5e-142	480.3	0.6	1.0	1	sp Q9SI17 PER14_ARATH	Peroxidase 14 OS=Arabidopsis thaliana
1.6e-142	480.3	4.0	1.8e-142	480.1	4.0	1.0	1	sp Q9FLC0 PER52_ARATH	Peroxidase 52 OS=Arabidopsis thaliana

Démarche : Construction de profils HMM



HMM contre Fichier Peroxybase

HMM : outil efficace

```
# hmmsearch :: search profile(s) against a sequence database
# HMMER 3.2.1 (June 2018); http://hmmer.org/
# Copyright (C) 2018 Howard Hughes Medical Institute.
# Freely distributed under the BSD open source license.
# -----
# query HMM file:          PB_APx_region_225_328.hmm
# target sequence database: /home/fmt017/work/projet_bioseq/séquences/peroxybase.fa
# output directed to file: PB_APx_region_225_328_0.search
# -----
```

Classe I
225-328

Query: PB_APx_region_225_328 [M=526]									
Scores for complete sequences (score includes all domains):									
--- full sequence ---			--- best 1 domain ---			-#dom-			
E-value	score	bias	E-value	score	bias	exp	N	Sequence	Description
4e-142	473.0	0.0	5.3e-142	472.6	0.0	1.0	1	2494 McAPx05	[Mesembryanthemum_crystallinum]
2e-141	470.8	0.0	2.1e-141	470.6	0.0	1.0	1	2608 AcAPx04	[Allium_cepa] Monocotyledons Asc
4.7e-141	469.5	0.0	5.1e-141	469.4	0.0	1.0	1	2153 HvAPx07	[Hordeum_vulgare] Monocotyledons
5.5e-141	469.3	0.0	7.1e-141	468.9	0.0	1.0	1	2879 PtAPx08	[Populus_trichocarpa] Eudicotyle
7.5e-141	468.8	0.0	8.2e-141	468.7	0.0	1.0	1	3426 CsAPx06	[Citrus_sinensis] Eudicotyledons
1.1e-140	468.3	0.0	1.2e-140	468.2	0.0	1.0	1	2232 GmAPx07	[Glycine_max] Eudicotyledons Asc
1.3e-140	468.0	0.0	1.6e-140	467.8	0.0	1.0	1	5136 MtAPx07-b	[Medicago_truncatula] Eudicotyle
4e-140	466.4	0.0	5.5e-140	466.0	0.0	1.2	1	1872 OsAPx08	[Oryza_sativa ssp japonica cv Ni
4.8e-140	466.2	0.0	5.7e-140	465.9	0.0	1.0	1	3956 VuAPx03-b	[Vigna_unguiculata] Eudicotyledo
5.9e-140	465.9	0.0	6.8e-140	465.7	0.0	1.0	1	2458 NtAPx01-b	[Nicotiana_tabacum] Eudicotyledo
1.2e-139	464.8	0.0	1.3e-139	464.7	0.0	1.0	1	2591 TaAPx01-1	[Triticum_aestivum] Monocotyledo
1.3e-139	464.7	0.0	1.5e-139	464.6	0.0	1.0	1	6711 ZmAPx08	[Zea_mays] Monocotyledons Ascorb
1.7e-139	464.4	0.0	1.9e-139	464.2	0.0	1.0	1	3278 AfpAPx03	[Aquilegia_formosa x Aquilegia p
1.9e-139	464.2	0.0	2e-139	464.1	0.0	1.0	1	2467 SoAPx01-b	[Spinacia_oleracea] Eudicotyledo
2.3e-139	463.9	0.0	2.6e-139	463.8	0.0	1.0	1	5172 VvAPx06	[Vitis_vinifera] Eudicotyledons
3e-139	463.5	0.0	3.8e-139	463.2	0.0	1.0	1	1895 BoAPx04	[Brassica_oleracea] Eudicotyledo
4e-139	463.2	0.0	4.5e-139	463.0	0.0	1.0	1	3503 SbAPx08	[Sorghum_bicolor] Monocotyledons
4e-139	463.1	0.0	5.1e-139	462.8	0.0	1.1	1	3962 TaAPx01_C306	[Triticum_aestivum] Monocotyledo

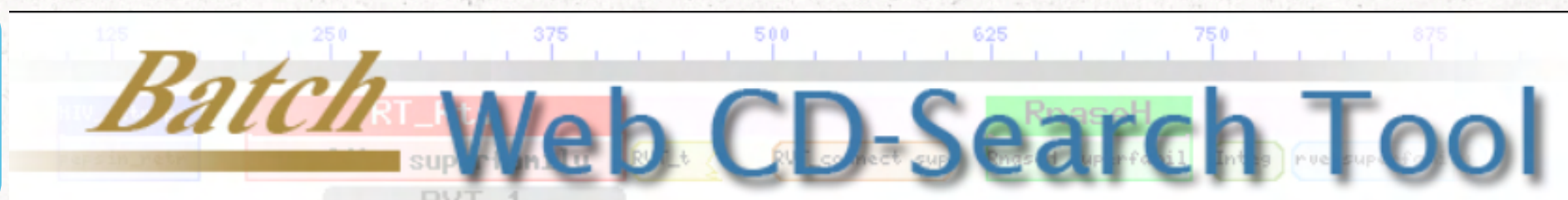
```
# hmmsearch :: search profile(s) against a sequence database
# HMMER 3.2.1 (June 2018); http://hmmer.org/
# Copyright (C) 2018 Howard Hughes Medical Institute.
# Freely distributed under the BSD open source license.
# -----
# query HMM file:          PB_classIII_conserved_region_110_290.hmm
# target sequence database: /home/fmt017/work/projet_bioseq/séquences/peroxybase.fa
# output directed to file: PB_classIII_conserved_region_110_290_0.search
# -----
```

Classe III
110-290

Query: PB_classIII_conserved_region_110_290 [M=436]									
Scores for complete sequences (score includes all domains):									
--- full sequence ---			--- best 1 domain ---			-#dom-			
E-value	score	bias	E-value	score	bias	exp	N	Sequence	Description
2.4e-150	499.8	0.0	2.6e-150	499.6	0.0	1.0	1	874 SbPrx71	[Sorghum_bicolor] Monocotyledons
1.1e-149	497.6	0.0	1.4e-149	497.2	0.0	1.0	1	7150 SmPrx100a_175	[Selaginella_moellendorffii] Cry
1.8e-149	496.9	0.9	2e-149	496.7	0.9	1.0	1	3080 BrPrx49-2	[Brassica_rapa] Eudicotyledons C
2.2e-149	496.6	0.7	2.4e-149	496.5	0.7	1.0	1	3053 BrPrx49-1	[Brassica_rapa] Eudicotyledons C
5.5e-149	495.3	1.1	6.1e-149	495.1	1.1	1.0	1	215 AtPrx49	[Arabidopsis_thaliana] Eudicotyl
6e-149	495.1	0.1	6.6e-149	495.0	0.1	1.0	1	802 SbPrx04	[Sorghum_bicolor] Monocotyledons
6.6e-149	495.0	1.4	7.8e-149	494.8	1.4	1.0	1	339 MtPrx12	[Medicago_truncatula] Eudicotyle
9.8e-149	494.4	0.6	1.1e-148	494.3	0.6	1.0	1	3108 PtPrx30	[Populus_trichocarpa] Eudicotyle
1.1e-148	494.2	0.5	1.2e-148	494.1	0.5	1.0	1	186 SoPrx15	[Spinacia_oleracea] Eudicotyledo
1.2e-148	494.1	0.0	1.3e-148	494.0	0.0	1.0	1	711 NtPrx05	[Nicotiana_tabacum] Eudicotyledo
1.2e-148	494.1	3.0	1.3e-148	494.0	3.0	1.0	1	2050 PtaPrx14	[Pinus_taeda] Gymnospermae Class
1.5e-148	493.9	1.2	1.6e-148	493.7	1.2	1.0	1	3454 LsaPrx08	[Lactuca_sativa] Eudicotyledons
1.9e-148	493.5	0.6	2.1e-148	493.4	0.6	1.0	1	370 MtPrx46	[Medicago_truncatula] Eudicotyle
1.9e-148	493.5	0.1	2.1e-148	493.4	0.1	1.0	1	714 ZmPrx18	[Zea_mays] Monocotyledons Class_
2e-148	493.4	1.6	2.4e-148	493.2	1.6	1.0	1	157 MsPrx02	[Medicago_sativa] Eudicotyledons
2.1e-148	493.3	1.9	2.4e-148	493.2	1.9	1.0	1	7062 SmPrx75b_233	[Selaginella_moellendorffii] Cry
2.6e-148	493.0	1.1	3.2e-148	492.8	1.1	1.0	1	439 GhPrx22	[Gossypium_hirsutum] Eudicotyled
2.8e-148	492.9	3.3	3.1e-148	492.8	3.3	1.0	1	1950 StPrx47	[Solanum_tuberosum] Eudicotyledo
3e-148	492.8	2.2	3.3e-148	492.7	2.2	1.0	1	2161 AfpPrx09	[Aquilegia_formosa x Aquilegia p
3.3e-148	492.7	0.0	3.6e-148	492.6	0.0	1.0	1	5111 VvPrx28	[Vitis_vinifera] Eudicotyledons
3.4e-148	492.7	0.0	3.8e-148	492.5	0.0	1.0	1	1854 SofPrx04	[Saccharum_officinarum] Monocoty

Démarche : CDSearch et Signal P

04



Query	Hit type	PSSM-ID	From	To	E-Value	Bitscore	Accession	Short name	Incomplete	Superfamily
Q#1 - >6508 MpPrx12	specific	173827	26	323	1.3394e-148	419.611	cd00693	secretory_peroxidase	-	cl00196
Q#2 - >6509 MpPrx13	specific	173827	51	340	2.39957e-132	380.321	cd00693	secretory_peroxidase	-	cl00196
Q#3 - >6511 MpPrx15	specific	173827	38	332	5.08471e-148	418.456	cd00693	secretory_peroxidase	-	cl00196
Q#4 - >7342 MpPrx16	specific	173827	28	326	1.23682e-127	366.454	cd00693	secretory_peroxidase	-	cl00196
Q#5 - >7343 MpPrx17	specific	173827	22	323	1.08982e-146	414.604	cd00693	secretory_peroxidase	-	cl00196
Q#6 - >250 PpaPrx10	specific	173827	38	327	3.94387e-163	456.59	cd00693	secretory_peroxidase	-	cl00196
Q#7 - >251 PpaPrx11	specific	173827	38	340	3.67439e-147	416.53	cd00693	secretory_peroxidase	-	cl00196
Q#8 - >3327 PpaPrx12	specific	173827	29	318	1.45918e-109	320.23	cd00693	secretory_peroxidase	-	cl00196
Q#9 - >3419 PpaPrx13	specific	173827	36	334	5.67078e-147	415.759	cd00693	secretory_peroxidase	-	cl00196
Q#10 - >3420 PpaPrx14	specific	173827	43	331	4.38716e-137	391.106	cd00693	secretory_peroxidase	-	cl00196

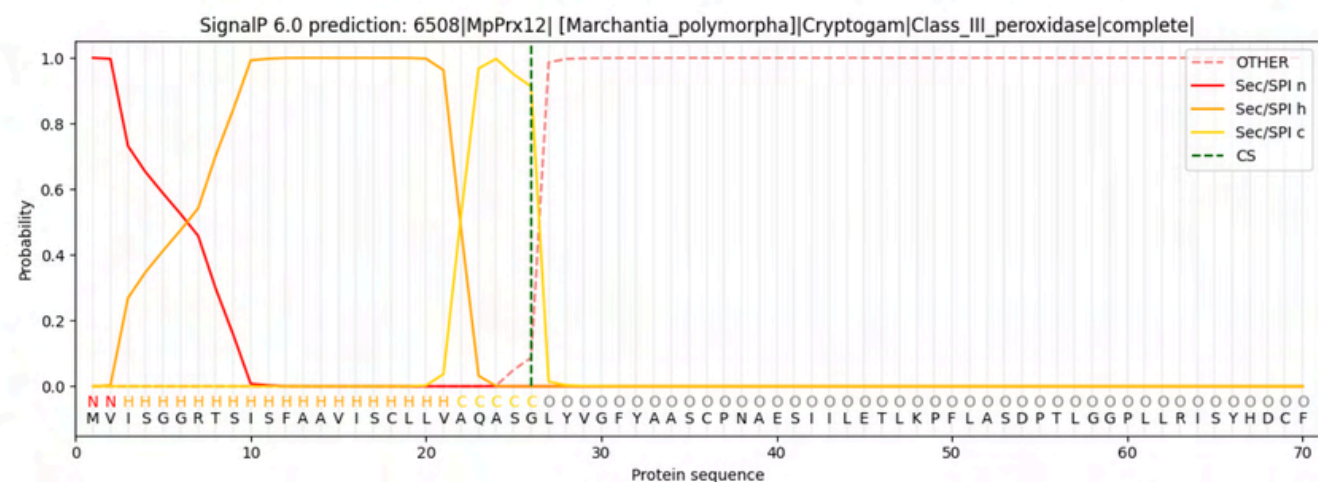
6508|MpPrx12| [Marchantia_polymorpha][Cryptogam][Class_III_peroxidase][complete]

Prediction: Signal Peptide (Sec/SPI)

Cleavage site between pos. 26 and 27. Probability 0.912813

Protein type	Other	Signal Peptide (Sec/SPI)
Likelihood	0.0005	0.9994

**Download: ** [PNG](#) / [EPS](#) / [Tabular](#)

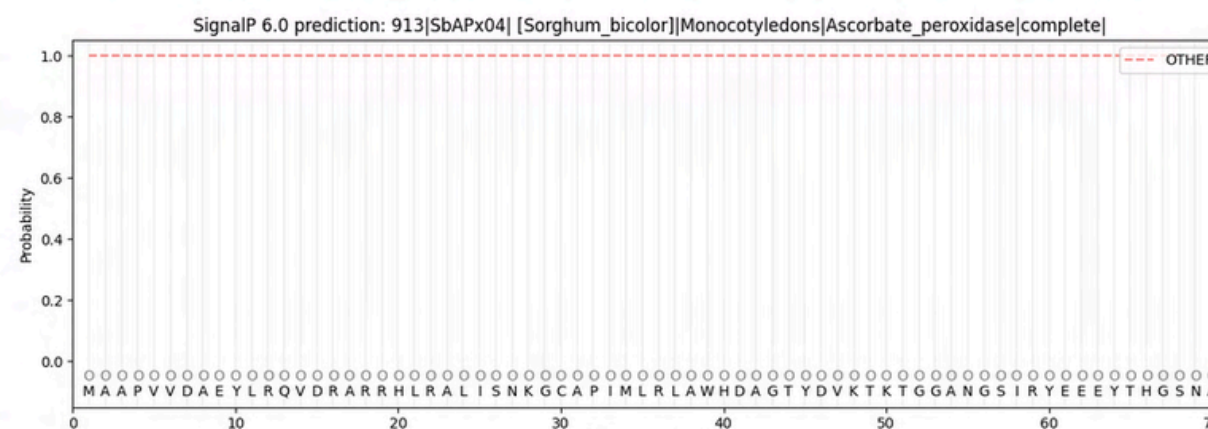


913|SbAPx04| [Sorghum_bicolor][Monocotyledons][Ascorbate_peroxidase][complete]

Prediction: Other

Protein type	Other	Signal Peptide (Sec/SPI)
Likelihood	1.0001	0

**Download: ** [PNG](#) / [EPS](#) / [Tabular](#)



Query	Hit type	PSSM-ID	From	To	E-Value	Bitscore	Accession	Short name	Incomplete	Superfamily
Q#1 - >376 KfIAPx03	superfamily	444739	96	327	1.71415e-49	165.455	cl00196	plant_peroxidase_like superfamily	-	-
Q#2 - >913 SbAPx04	specific	178218	1	289	0	575.945	PLN02608	PLN02608	-	cl29002
Q#3 - >995 ZmAPx05	superfamily	444739	1	250	2.71762e-158	439.131	cl00196	plant_peroxidase_like superfamily	-	-
Q#4 - >1383 CsAPx01	superfamily	444739	1	250	1.56833e-150	419.486	cl00196	plant_peroxidase_like superfamily	-	-
Q#5 - >1650 McAPx03	specific	173825	3	244	1.75187e-142	398.885	cd00691	ascorbate_peroxidase	-	cl00196
Q#6 - >1670 CrAPx01	superfamily	178218	1	273	0	504.683	cl29002	PLN02608 superfamily	-	-
Q#7 - >1671 CrAPx02	specific	173825	5	249	6.73775e-151	420.842	cd00691	ascorbate_peroxidase	-	cl00196
Q#8 - >1707 PbAPx01	superfamily	444739	1	248	1.72528e-160	444.888	cl00196	plant_peroxidase_like superfamily	-	-
Q#9 - >1710 PbAPx02	superfamily	444739	1	248	1.36096e-157	437.57	cl00196	plant_peroxidase_like superfamily	-	-
Q#10 - >1755 GhyAPx01	specific	173825	5	250	2.92488e-150	418.916	cd00691	ascorbate_peroxidase	-	cl00196

Concrétisation de notre démarche

Find your protein

UniProtKB ▾ Peroxidase

Advanced | List Search

Examples: Insulin, APP, Human, P05067, organism_id:9606

Popular organisms

A. thaliana (567)



Annotation score

1 (64)



FASTA

```
hmmsearch :: search profile(s) against a sequence database
HMMER 3.2.1 (June 2018); http://hmmer.org/
Copyright (C) 2018 Howard Hughes Medical Institute.
Freely distributed under the BSD open source license.
```

```
Query HMM file:          PB_APx_region_225_328.hmm
Target sequence database: /home/fmt017/work/projet_bioseq/test2_px.fasta
Output directed to file:  PB_APx_region_225_328_test2_px.search
```

Query: PB_APx_region_225_328 [M=526]

Results for complete sequences (score includes all domains):

--- full sequence ---			--- best 1 domain ---			-#dom-		Sequence	Description
E-value	score	bias	E-value	score	bias	exp	N		
4.9e-61	198.8	0.0	2.4e-50	163.5	0.0	2.0	2	tr A0A1P8B8Y3 A0A1P8B8Y3_ARATH	Ascorbate peroxidase 4 OS=Arabid
8.2e-61	198.0	0.0	6.9e-58	188.4	0.0	2.0	1	tr A0A5S9XSG0 A0A5S9XSG0_ARATH	(thale cress) hypothetical prote
2.2e-60	196.6	0.0	1e-49	161.4	0.0	2.0	2	tr A0A654FMI0 A0A654FMI0_ARATH	Plant heme peroxidase family pro
2.4e-60	196.5	0.0	1.1e-49	161.3	0.0	2.0	2	tr A0A178UTM3 A0A178UTM3_ARATH	TL29 OS=Arabidopsis thaliana OX=
4.8e-60	195.5	0.0	3.2e-49	159.8	0.0	2.0	2	tr A0A1P8B8W6 A0A1P8B8W6_ARATH	Ascorbate peroxidase 4 OS=Arabid
4.5e-31	99.9	0.0	7.2e-18	56.3	0.0	2.0	2	tr A0A5S9WLA6 A0A5S9WLA6_ARATH	Uncharacterized protein OS=Arabi
1.5e-14	45.3	0.0	1.8e-14	45.1	0.0	1.0	1	tr Q0WQ25 Q0WQ25_ARATH	Uncharacterized protein OS=Arabi
3.1e-09	27.8	0.0	3.1e-09	27.8	0.0	1.0	1	tr Q56ZQ1 Q56ZQ1_ARATH	peroxidase OS=Arabidopsis thalia
5.6e-05	13.8	0.0	0.0001	12.9	0.0	1.4	1	tr A0A7G2FKM8 A0A7G2FKM8_ARATH	(thale cress) hypothetical prote

Classe I
225-328

```
hmmsearch :: search profile(s) against a sequence database
HMMER 3.2.1 (June 2018); http://hmmer.org/
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```

```
Query HMM file:          PB_classIII_conserved_region_110_290.hmm
Target sequence database: test2_px.fasta
Output directed to file:  PB_classIII_conserved_region_110_290_test2_px.search
```

Query: PB_classIII_conserved_region_110_290 [M=436]

Results for complete sequences (score includes all domains):

--- full sequence ---			--- best 1 domain ---			-#dom-		Sequence	Description
E-value	score	bias	E-value	score	bias	exp	N		
6.4e-19	60.0	0.1	6.8e-19	59.9	0.1	1.0	1	tr Q56ZQ1 Q56ZQ1_ARATH	peroxidase OS=Arabidopsis thalia
1.4e-17	55.5	0.0	1.6e-17	55.4	0.0	1.0	1	tr A0A178UTE8 A0A178UTE8_ARATH	Plant heme peroxidase family pro
3e-11	34.7	0.0	6e-10	30.4	0.0	2.2	2	tr A0A1P8B8W6 A0A1P8B8W6_ARATH	Ascorbate peroxidase 4 OS=Arabid
3.9e-11	34.3	0.1	5.6e-10	30.5	0.0	2.2	2	tr A0A654FMI0 A0A654FMI0_ARATH	Plant heme peroxidase family pro
3.9e-11	34.3	0.0	8.3e-10	29.9	0.0	2.2	2	tr A0A178UTM3 A0A178UTM3_ARATH	TL29 OS=Arabidopsis thaliana OX=
5.3e-11	33.9	0.0	1.1e-09	29.5	0.0	2.2	2	tr A0A5S9XSG0 A0A5S9XSG0_ARATH	(thale cress) hypothetical prote
8.2e-11	33.2	0.1	1.4e-09	29.2	0.0	2.2	2	tr A0A1P8B8Y3 A0A1P8B8Y3_ARATH	Ascorbate peroxidase 4 OS=Arabid
2e-08	25.4	0.1	3.5e-08	24.6	0.1	1.3	1	tr A0A7G2FKM8 A0A7G2FKM8_ARATH	(thale cress) hypothetical prote
3.4e-07	21.3	0.0	5.1e-07	20.8	0.0	1.2	1	tr A0A5S9WLA6 A0A5S9WLA6_ARATH	Uncharacterized protein OS=Arabi

Classe III
110-290

Concrétisation de notre démarche

HMM contre séquences non caractérisées de la sphaigne

```
# HMMER 3.2.1 (June 2018); http://hmmer.org/
# Copyright (C) 2018 Howard Hughes Medical Institute.
# Freely distributed under the BSD open source license.
# -----
# query HMM file:          PB_classIII_conserved_region_110_290.hmm
# target sequence database: /home/fmt017/work/projet_bioseq/séquences/sphaigne.fasta
# output directed to file: PB_classIII_conserved_region_110_290_vs_sphaigne.search
# -----

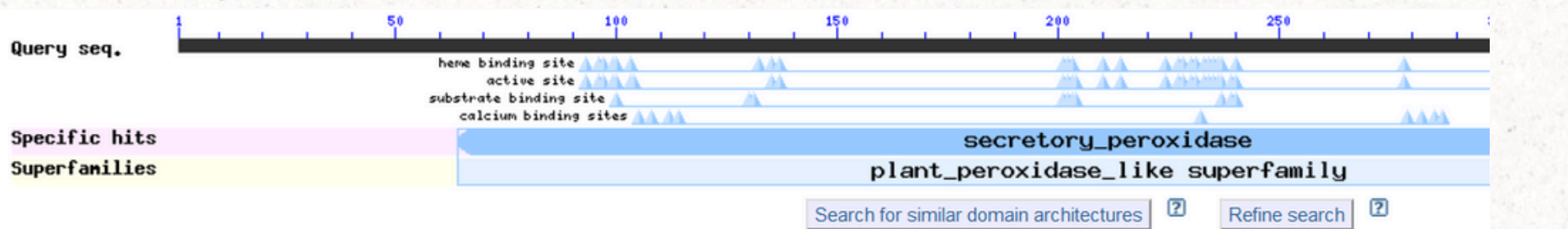
Query:      PB_classIII_conserved_region_110_290 [M=436]
Scores for complete sequences (score includes all domains):
--- full sequence ---  --- best 1 domain ---  -#dom-
  E-value  score  bias  E-value  score  bias  exp  N  Sequence              Description
-----
1.6e-136  448.0   0.2  1.8e-136  447.8   0.2   1.0  1  Sphfalx06G076300|Sphfalx06G076300.1
2.6e-135  444.0   0.0  2.7e-135  444.0   0.0   1.0  1  Sphfalx14G080900|Sphfalx14G080900.1
5.9e-135  442.8   0.5  6.9e-135  442.6   0.5   1.0  1  Sphfalx02G005500|Sphfalx02G005500.1
1.3e-134  441.7   1.4  1.5e-134  441.5   1.4   1.0  1  Sphfalx08G109900|Sphfalx08G109900.1
1.8e-134  441.3   0.1  2e-134   441.1   0.1   1.0  1  Sphfalx13G084900|Sphfalx13G084900.1
2.3e-134  440.9   0.2  2.5e-134  440.8   0.2   1.0  1  Sphfalx03G096500|Sphfalx03G096500.1
5.8e-134  439.6   0.1  6.8e-134  439.3   0.1   1.0  1  Sphfalx07G098400|Sphfalx07G098400.1
9.8e-134  438.8   0.2  1.1e-133  438.7   0.2   1.0  1  Sphfalx03G096600|Sphfalx03G096600.1
1.2e-133  438.6   0.5  1.4e-133  438.3   0.5   1.0  1  Sphfalx07G023500|Sphfalx07G023500.1
```

Classe III
110-290

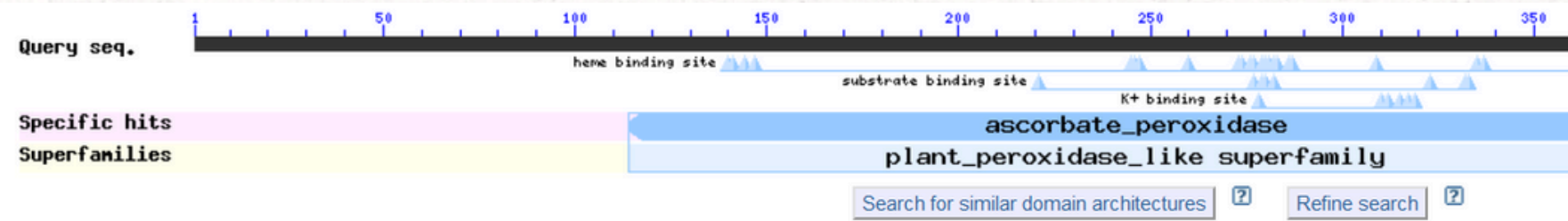
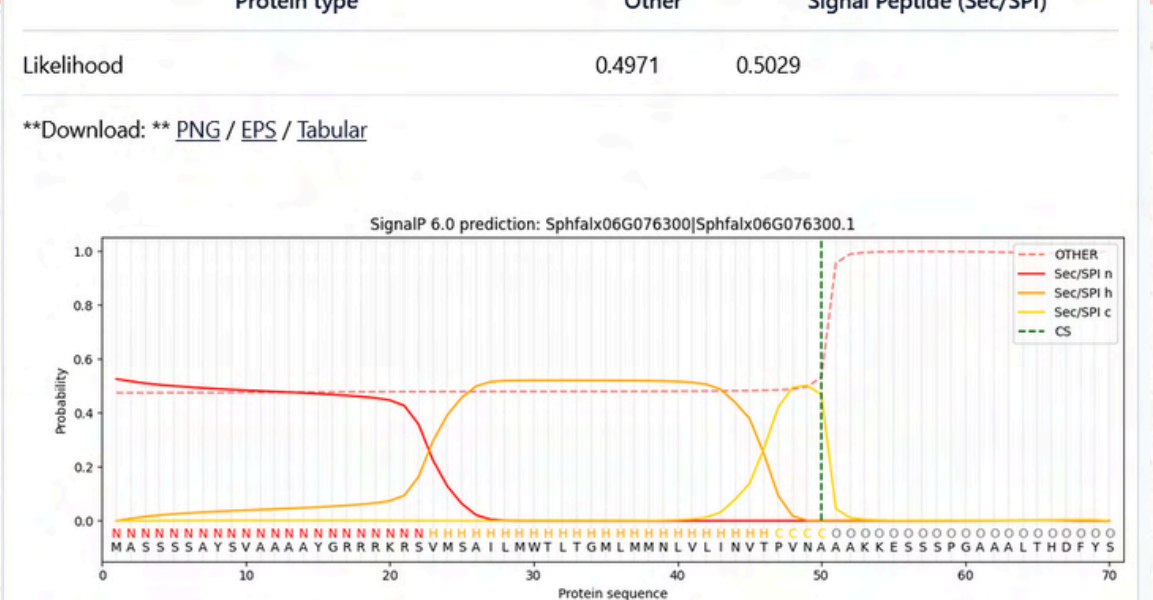
```
# hmmsearch :: search profile(s) against a sequence database
# HMMER 3.2.1 (June 2018); http://hmmer.org/
# Copyright (C) 2018 Howard Hughes Medical Institute.
# Freely distributed under the BSD open source license.
# -----
# query HMM file:          PB_APx_region_225_328.hmm
# target sequence database: /home/fmt017/work/projet_bioseq/séquences/sphaigne.fasta
# output directed to file: PB_APx_region_225_328_vs_sphaigne.search
# -----

Query:      PB_APx_region_225_328 [M=526]
Scores for complete sequences (score includes all domains):
--- full sequence ---  --- best 1 domain ---  -#dom-
  E-value  score  bias  E-value  score  bias  exp  N  Sequence              Description
-----
7.1e-139  456.1   0.0  8.8e-139  455.8   0.0   1.0  1  Sphfalx03G024900|Sphfalx03G024900.1
2.1e-136  448.0   0.0  2.5e-136  447.7   0.0   1.0  1  Sphfalx17G035500|Sphfalx17G035500.1
2.1e-136  448.0   0.0  2.5e-136  447.7   0.0   1.0  1  Sphfalx17G035500|Sphfalx17G035500.2
9.3e-132  432.6   0.0  1e-131   432.5   0.0   1.0  1  Sphfalx05G037200|Sphfalx05G037200.1
1.3e-128  422.3   0.0  1.5e-128  422.1   0.0   1.0  1  Sphfalx12G091800|Sphfalx12G091800.1
1.8e-128  421.8   0.0  2.1e-128  421.6   0.0   1.0  1  Sphfalx12G091800|Sphfalx12G091800.2
1.8e-128  421.8   0.0  2.1e-128  421.6   0.0   1.0  1  Sphfalx12G091800|Sphfalx12G091800.3
4.3e-126  414.0   0.0  4.7e-126  413.8   0.0   1.0  1  Sphfalx07G109700|Sphfalx07G109700.1
1.3e-122  402.5   0.0  1.5e-122  402.2   0.0   1.0  1  Sphfalx12G068700|Sphfalx12G068700.1
```

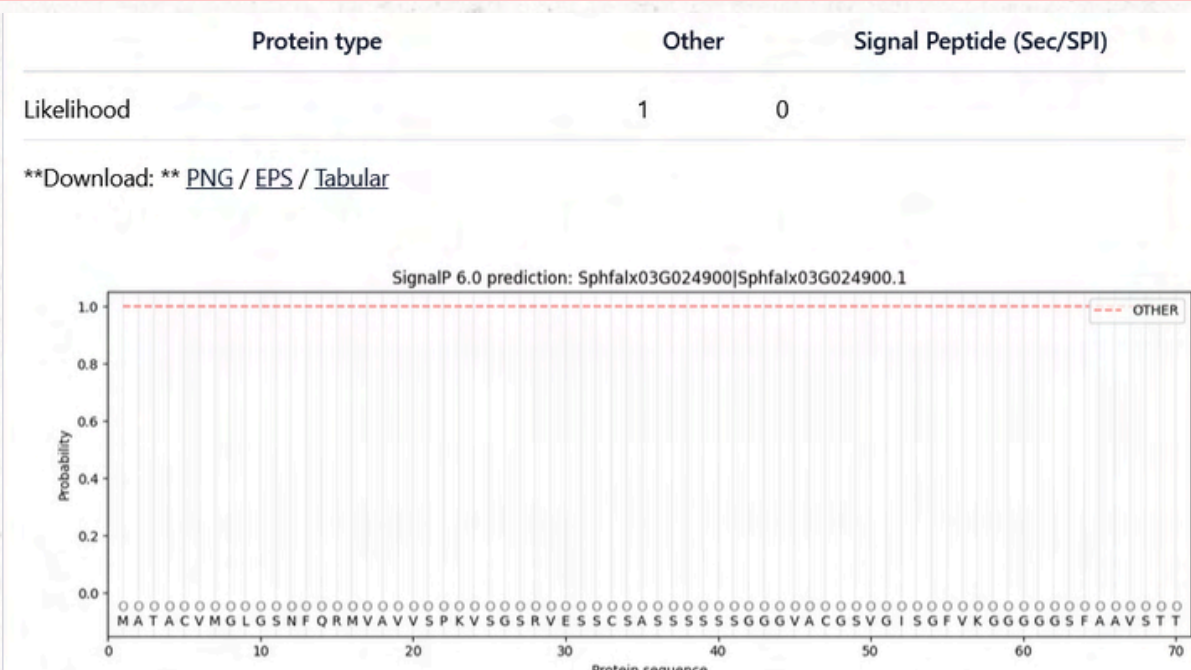
Classe I
225-328



List of domain hits				
	Name	Accession	Description	E-value
+	secretory_peroxidase	cd00693	Horseradish peroxidase and related secretory plant peroxidases; Secretory peroxidases belong ...	3.12e-160



List of domain hits				
	Name	Accession	Description	E-value
+	ascorbate_peroxidase	cd00691	Ascorbate peroxidases and cytochrome C peroxidases; Ascorbate peroxidases are a subgroup of ...	5.74



Conclusion

Utilisation des outils identifiés pour caractériser une séquence inconnue

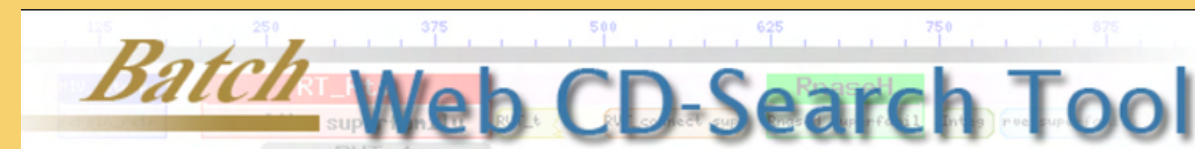
Séquence protéique non classifiée ou mal annotée



Séquence contre
-HMM domaine conservé classe I
-HMM domaine conservé classe III



Vérifications complémentaires



THANK YOU!

GN
D
F
P
Y
D
P
C
P
T
P